

A Minimal Copper-Plate Storage-Scheduling Benchmark for Differential Dynamic Programming

Aryan Ritwajeet Jha, *Student Member, IEEE*

I. COPPER-PLATE STORAGE-SCHEDULING MODEL

A. Optimization Problem

A single storage device and a single grid connection serve a known demand p_L^t over T periods of length Δt , all quantities in per unit. Following the sign convention of [1], $P_B^t > 0$ discharges and $P_B^t < 0$ charges; B^t is the energy at the *start* of period t , so B^1 is given and B^{T+1} is terminal. With $t \in \mathcal{T} = \{1, \dots, T\}$:

$$\min_{P_{\text{Subs}}, P_B, B} \sum_{t=1}^T \left[c^t P_{\text{Subs}}^t + C_B (P_B^t)^2 \right] \Delta t + w (B^{T+1} - B^1)^2 \quad (1a)$$

$$\text{s.t. } P_{\text{Subs}}^t + P_B^t - p_L^t = 0, \quad t \in \mathcal{T} \quad (1b)$$

$$B^1 = B_0 \quad (1c)$$

$$B^{t+1} = B^t - P_B^t \Delta t, \quad t \in \mathcal{T} \quad (1d)$$

$$P_{\text{Subs}}^t \geq 0, \quad t \in \mathcal{T} \quad (1e)$$

$$P_B^{\min} \leq P_B^t \leq P_B^{\max}, \quad t \in \mathcal{T} \quad (1f)$$

$$B^{\min} \leq B^{t+1} \leq B^{\max}, \quad t \in \mathcal{T} \quad (1g)$$

The *base instance* is (1a)–(1d), with all three bounds dropped.

B. Optimal-Control Form

Restated over $N = T$ stages in the form required by DDP:

$$\begin{aligned} \min_{x, u} \quad & \sum_{t=1}^{N-1} \ell(x^t, u^t) + \ell^F(x^N, u^N) \\ \text{s.t. } \quad & x^1 = \hat{x}^1, \\ & x^{t+1} = f(x^t, u^t), \quad t \in \{1, \dots, N-1\}, \\ & c(x^t, u^t) = 0, \quad t \in \{1, \dots, N\}. \end{aligned} \quad (2)$$

with state, control and initial condition

$$x^t = (B^t, \tau^t)^\top \in \mathbb{R}^2, \quad u^t = (P_{\text{Subs}}^t, P_B^t)^\top \in \mathbb{R}^2, \quad (3)$$

$\hat{x}^1 = (B_0, 1)^\top$. Period-varying data is carried by the time index τ through the Lagrange interpolants on the nodes $\tau = 1, \dots, T$,

$$p_L(\tau) = \sum_{k=1}^T p_L^k \prod_{\substack{m=1 \\ m \neq k}}^T \frac{\tau - m}{k - m}, \quad (4)$$

and likewise for the price $c(\tau)$. The stage data is then

$$f(x, u) = (x_1 - u_2 \Delta t, x_2 + 1)^\top, \quad (5)$$

$$c(x, u) = u_1 + u_2 - p_L(x_2), \quad (6)$$

$$\ell(x, u) = [c(x_2) u_1 + C_B u_2^2] \Delta t, \quad (7)$$

$$\begin{aligned} \ell^F(x, u) = & [c^T u_1 + C_B u_2^2] \Delta t \\ & + w (x_1 - u_2 \Delta t - B_0)^2. \end{aligned} \quad (8)$$

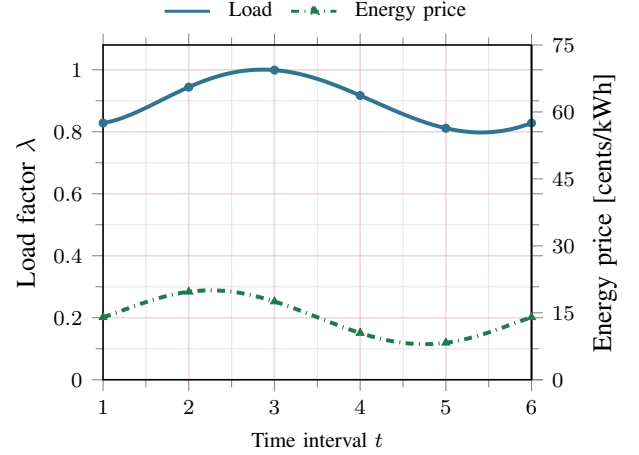


Fig. 1. Input profiles over the $T = 6$ horizon: load factor on the left axis, energy price on the right, each specified at the T horizon points. Markers are the data; the curves are the Lagrange interpolants (4) through them, and it is the curves, not the markers, that the solver evaluates. The two agree exactly at every node, by construction, but between nodes they need not be well behaved and already are not — the load interpolant reaches 0.998 against a node maximum of 0.9, and the price interpolant 69.7 cents/kWh against 62. Nothing samples those excursions here, because the time-index state advances by exactly one per stage and so only ever lands on a node; but the interpolant's derivatives at the nodes are what the backward pass consumes, and its degree is $T - 1$. This is the device that does not survive a realistic horizon.

Bounds (1e)–(1f) map onto control limits $b^L \leq u^t \leq b^U$. The state bound (1g) is appended as a second row of c via a slack s^t adjoined to the control, giving $n_c = 2$, $n_u = 3$:

$$(x_1 - u_2 \Delta t) - B^{\min} - s^t = 0, \quad 0 \leq s^t \leq B^{\max} - B^{\min}. \quad (9)$$

C. Closed-Form Reference

The base instance is a strictly convex equality-constrained QP. Eliminating $P_{\text{Subs}}^t = p_L^t - P_B^t$ and $B^{T+1} - B^1 = -\Delta t \sum_t P_B^t$ reduces it to

$$\begin{aligned} J(P_B) = & \text{const} - \Delta t \sum_t c^t P_B^t + C_B \Delta t \sum_t (P_B^t)^2 \\ & + w (\Delta t)^2 s^2, \end{aligned} \quad (10)$$

with $s = \sum_t P_B^t$. Stationarity divided through by Δt is $-c^k + 2C_B P_B^k + 2w \Delta t s = 0$; summing over k collapses it to

$$s = \frac{\sum_t c^t}{2C_B + 2Tw \Delta t}, \quad P_B^k = \frac{c^k - 2w \Delta t s}{2C_B}. \quad (11)$$

REFERENCES

- [1] A. R. Jha, S. Paul, and A. Dubey, "Analyzing the performance of linear and nonlinear multi-period optimal power flow models for active distribution networks," in *2025 IEEE North-East India International Energy Conversion Conference and Exhibition (NE-IECCCE)*. IEEE, 2025, pp. 04–06.

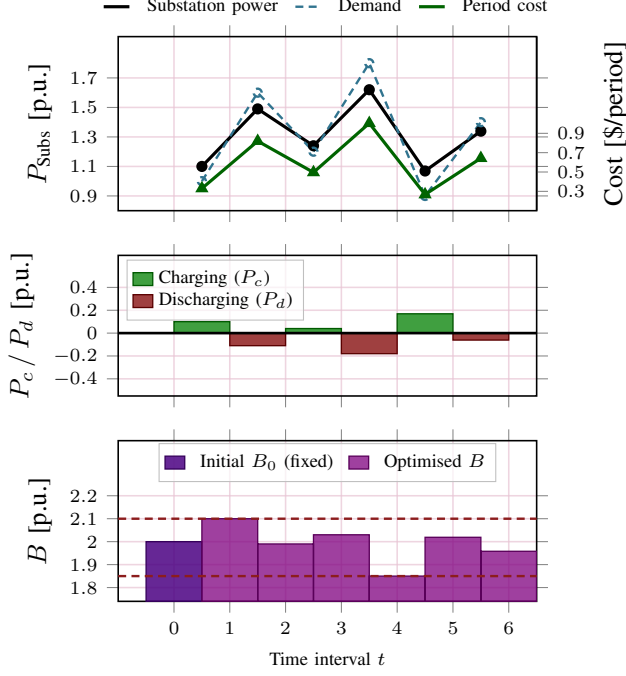


Fig. 2. The benchmark solved at $T = 6$ with all bounds (1e)–(1g) imposed. Battery actions use the convention of the tADMM study: charging is drawn upward in green and discharging downward in maroon, so a bar's direction reads as the device's action rather than as the sign of P_B^t . Read it as one worked example. The battery charges in the cheap intervals $t = 1, 3, 5$ and discharges into the expensive ones $t = 2, 4, 6$, which is why the substation trace in the top panel is visibly flatter than the demand it serves — it is pulled down at the two demand peaks and pushed up in the trough at $t = 5$, where energy is cheapest. Substation import is unlimited above, so the schedule is bounded only by the device: stored energy is pushed onto $B^{\max} = 2.10$ after the first charge and drawn right down onto $B^{\min} = 1.85$ after the interval-4 discharge (dashed rules in the lower panel). Those two limits are exactly the active set of the independent reference solver reports, and the trajectory sits on them to 1.2×10^{-9} . The battery power bounds, $-0.5 \leq P_B^t \leq 0.30$, never bind.

TABLE I
INSTANCE DATA

Interval t	1	2	3	4	5	6
<i>Horizon $T = 6$</i>						
Demand p_L^t	1.657	1.888	1.998	1.834	1.623	1.657
Energy price c^t	0.140	0.197	0.175	0.105	0.083	0.140
<i>Horizon $T = 3$ — resampled, not a prefix of the above</i>						
Demand p_L^t	1.657	1.943	1.657	—		
Energy price c^t	0.140	0.140	0.140	—		
Period length Δt	1	Initial energy B_0		2.0		
Throughput C_B	0.5	Terminal weight w		5.0		
<i>Bounds, constrained variants only</i>			$T = 3$		$T = 6$	
$P_B^{\min}, P_B^{\max}$			−0.5, 0.004		−0.045, 0.045	
$B^{\min}, B^{\max}$			1.9895, 1.9965		1.92, 1.99	

All quantities are in per unit; with $\Delta t = 1$, power and energy are numerically interchangeable. The series are the tADMM load and price profiles, $p_L^t = 2\lambda^t$ with $\lambda^t = 0.8 + 0.1(\sin(\theta^t - 0.8) + 1)$ and $c^t = 0.08 + 0.06(\sin \theta^t + 1)$, sampled at $\theta^t \in \text{range}(0, 2\pi, T)$. They therefore *resample* with T : the $T = 3$ row is not the first three entries of the $T = 6$ row. Two consequences of that sampling are worth stating plainly. First, both endpoints of $[0, 2\pi]$ are included, so $c^1 = c^T$ for every T . Second, at $T = 3$ the samples fall at $0, \pi$ and 2π , where $\sin$ vanishes, so *the price is constant* and the $T = 3$ instance carries no arbitrage signal at all — its optimal P_B^t is the same in every period and is set purely by the terminal penalty. The -0.8 phase offset on the load is what keeps the two series from being collinear; their Pearson correlation is 0.644 at $T = 6$. Bounds apply only to the constrained variants. Figs. 1 and 2 both show the $T = 6$ case.